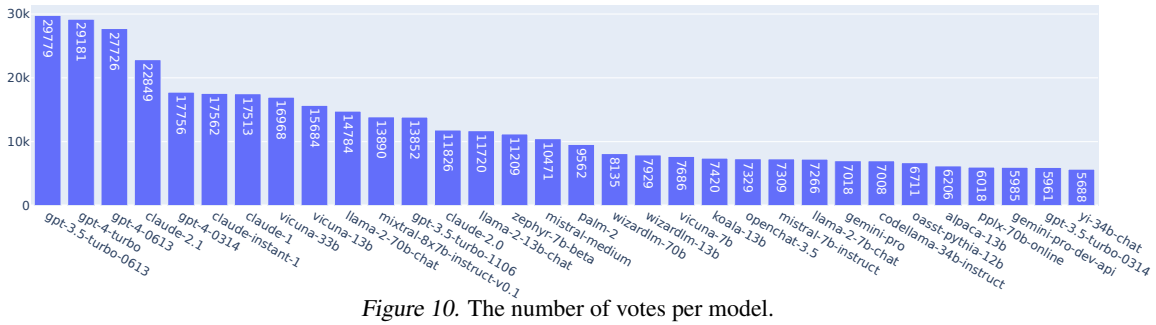

Figure 10. The number of votes per model.

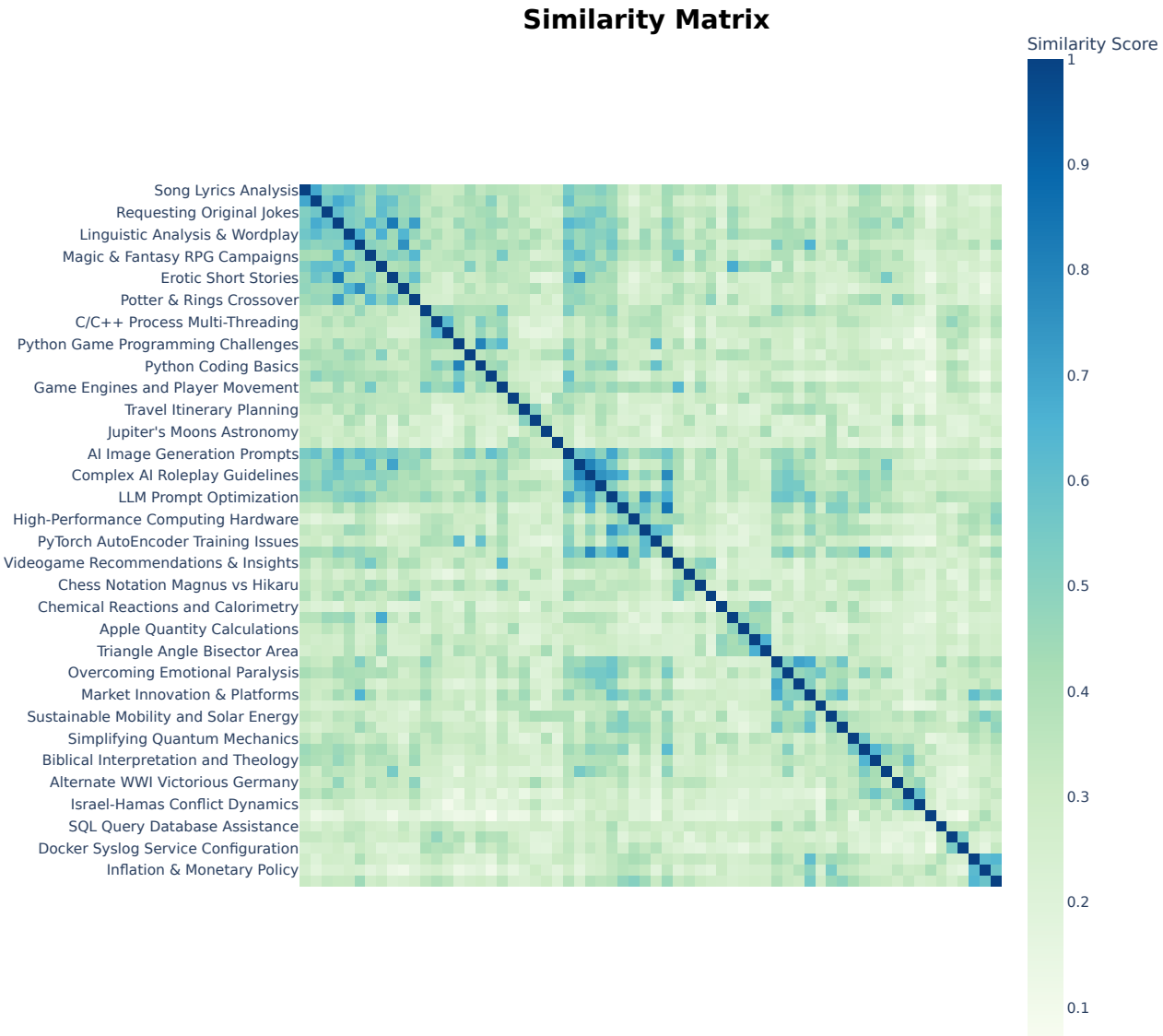

Figure 11. Similarity matrix of top-64 topic clusters.

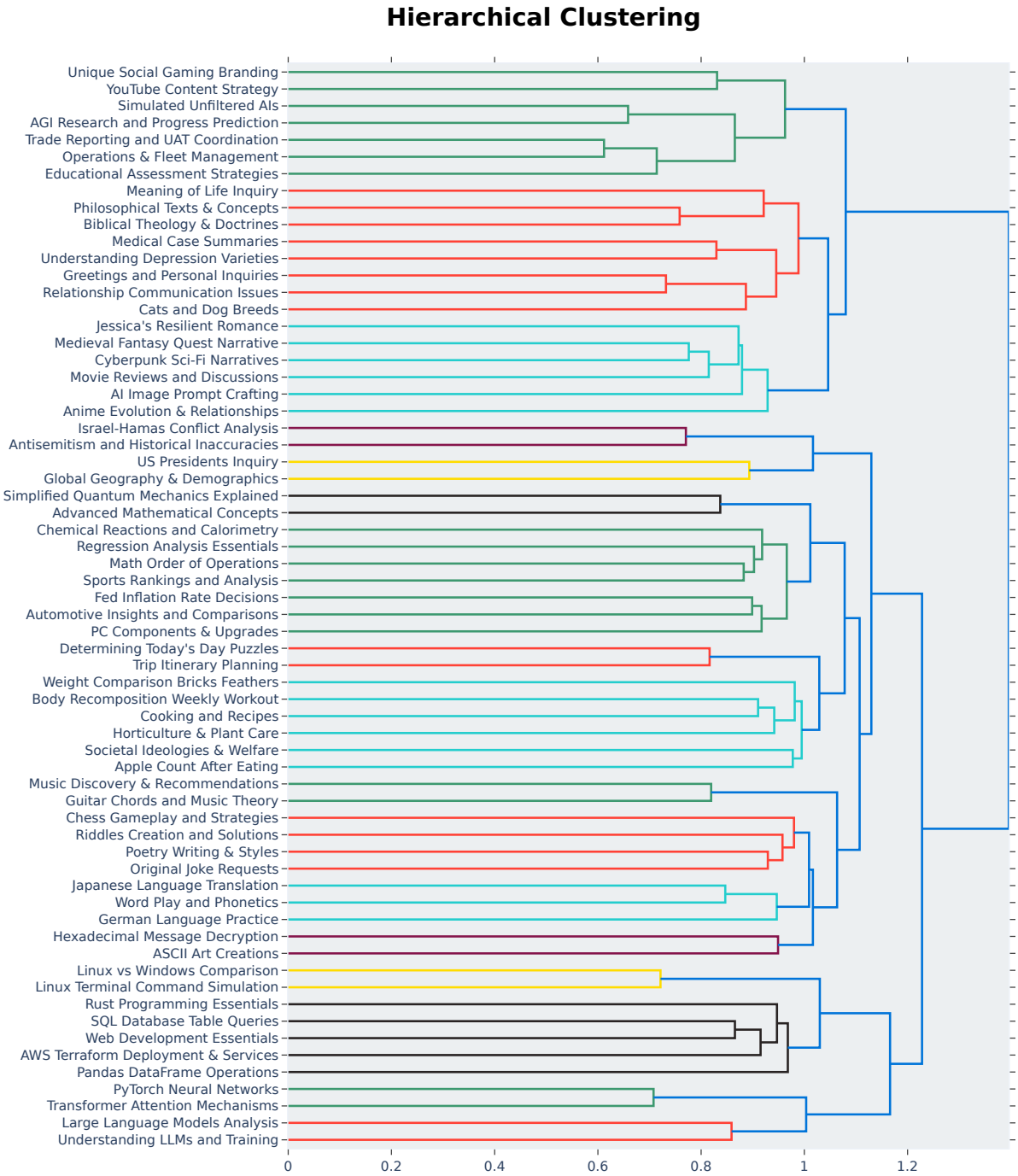

Figure 12. Top-64 clusters visualized in hierarchy. x-axis represents the cosine similarity distance. y-axis shows the topic title per cluster summarized by gpt-4-turbo.

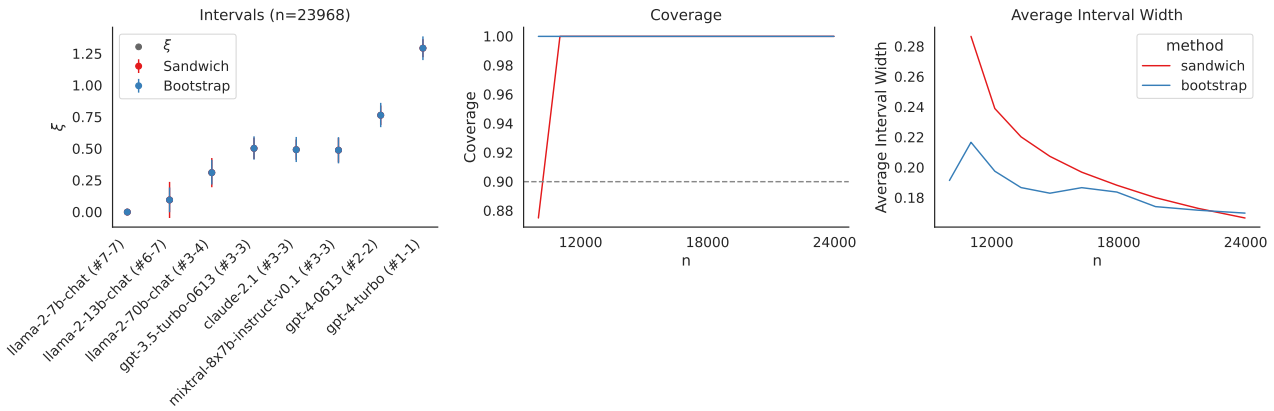

Figure 13. Replay experiment showing the intervals, coverage, and average interval sizes of the bootstrap and of the sandwich intervals. The sandwich intervals, though larger in small samples, are more stable, and in large samples, they actually become smaller. We use the multiplicity corrected version of both intervals, so they both have a coverage of 1. (Coverage here is calculated with respect to the BT coefficient solution on the full dataset, so it is not as meaningful as in the simulation plot below.)

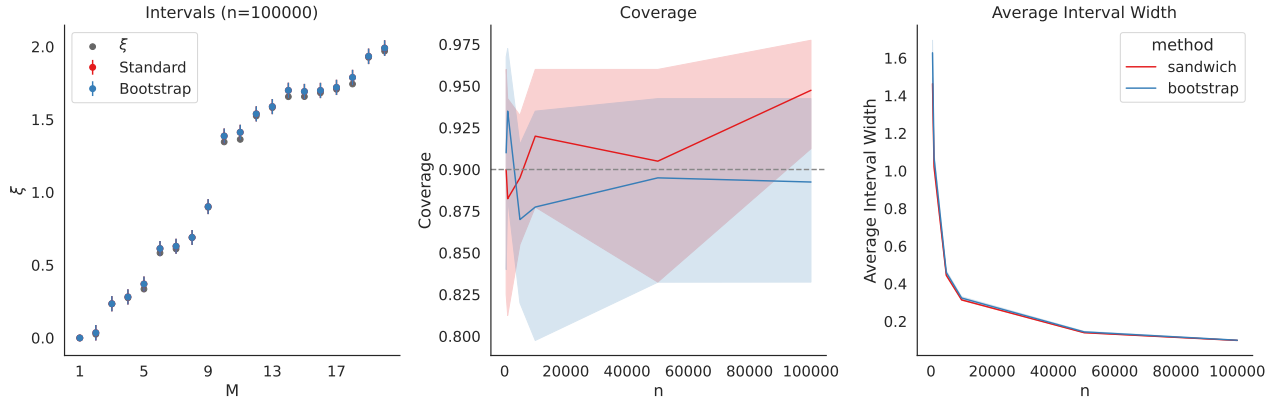

Figure 14. Synthetic experiment. Coefficients are drawn from the BT-coefficient distribution x on the left. Coverage of the uncorrected intervals is shown in the middle. Line plots of set width are shown on the right, and they almost perfectly match.

A. Confidence Interval Simulation Study

We conduct a simulation study to evaluate the bootstrap confidence intervals versus the sandwich estimator. To a large extent, both intervals are the same—indeed, their intervals are often identical to the naked eye. Nonetheless, in our experiments, there are some differences. First, in Figure 13, we conduct a replay study using the same 213576 data points mentioned in the main text.

We also do a suite of experiments in simulation using the same beta generating process as in the main text, with $\gamma = 2$. The result is shown in Figure 14; results are similar across many choices of the parameter γ and the model strength, which indicates that both intervals will have good coverage and width in the practical conditions we would expose them to.

B. The Nonparametric Bradley-Terry Model

Nonparametric Bradley-Terry. We next consider a nonparametric extension of the Bradley-Terry (BT) model (Bradley & Terry, 1952) to the case where the ranking is not necessarily transitive. Let $\mathcal{G}(m)$ denote the set of all *paths* to the model m, i.e.,

$$\mathcal{G}(m) = \left\{ g \in \mathcal{B}^{M-1} : g_{i,1} \neq g_{j,1},\ \forall i \neq j, \ \text{ and } g_{M-1,2} = m \right\}, \tag{11}$$

where $\mathcal{B} = \mathcal{A} \cup \{(a_2, a_1) : a \in \mathcal{A}\}$. Each element of $\mathcal{G}(m)$ is a chain of model pairings that leads to m; for example, if $m = 5$ and $M = 6$, one element of $\mathcal{G}(m)$ is $((1,2),(2,4),(4,3),(3,6),(6,5))$. Our score function is given by the average